

Question:

A baby is born after the 32nd gestational week by cesarean delivery. The mother suffered from gestational diabetes; however, she had no other pregnancy-related diseases and was otherwise healthy. The baby has a blood pressure of 100/58 mm Hg, heart rate of 104/min, and oxygen saturation of 88%. The child has tachypnea, subcostal and intercostal retractions, nasal flaring, and cyanosis. The cyanosis is responding well to initial administration of oxygen. The nasogastric tube was positioned without problems. Which of the following is the most likely diagnosis?

Answer Choices:

- A. Neonatal respiratory distress syndrome (NRDS)
- B. Tracheoesophageal fistula
- C. Pneumonia
- D. Congenital heart anomaly with right-to-left shunt

Reasoning + Answer:

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Finding Reasoning Paths :

1. **Neonatal respiratory distress syndrome (NRDS)**: NRDS is a common cause of tachypnea, subcostal and intercostal retractions, cyanosis, and hypoxia in preterm infants. It is often associated with respiratory insufficiency and requires oxygen supplementation.
2. **Persistent pulmonary hypertension (PPHN)**: PPHN can cause cyanosis and respiratory distress but typically presents with tachycardia and hypoxemia, not dyspnea.
3. **Severe upper airway obstruction**: Obstruction at the level of the trachea could lead to respiratory distress, cyanosis, and oxygen desaturation, but this would likely cause a barking cough or stridor rather than the presentation described.
4. **Chlamydial pneumonia**: Pneumonia, especially chlamydial pneumonia, can cause cyanosis, tachypnea, and respiratory distress, but it typically presents with fever and lower respiratory signs.
5. **Infantile asthma**: Asthma in neonates is rare and typically presents with wheezing and dyspnea, not the specific clinical findings described.

Reasoning Process :

1. **Initial Presentation**: The baby was born at 32 weeks' gestation, which places it in the category of a preterm infant. Preterm infants are at high risk for respiratory complications due to immature lung development. The symptoms of tachypnea, cyanosis, and oxygen desaturation are consistent with respiratory insufficiency.
2. **Cyanosis and Desaturation**: Cyanosis and oxygen desaturation in this setting strongly suggest hypoxemia. NRDS, caused by insufficient surfactant and alveolar collapse, is a common cause of respiratory distress and oxygen dependency in preterm infants.
3. **No Fever or Stridor**: The absence of fever, stridor, or abnormal cry distinguishes this case from conditions like chlamydial pneumonia or infantile asthma. These conditions are more likely to present with lower respiratory signs or wheezing.
4. **Resolution of Symptoms with Oxygen**: The improvement of symptoms with oxygen administration further supports a diagnosis of NRDS, as it is a common cause of respiratory distress in this population that responds to oxygen therapy.
5. **Exclusion of Other Diagnoses**:
 - **PPHN**: Typically presents with cyanosis, tachycardia, and hypoxemia, not the dyspnea described.
 - **Severe upper airway obstruction**: Would cause stridor or a barking cough, not the described presentation.
 - **Chlamydial pneumonia**: Would likely present with fever, lower respiratory signs, and pneumonia.
 - **Infantile asthma**: Would present with wheezing and dyspnea, not cyanosis and oxygen dependency.

Conclusion :

The most likely diagnosis is **neonatal respiratory distress syndrome (NRDS)**.

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<answer>

Neonatal respiratory distress syndrome (NRDS)

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